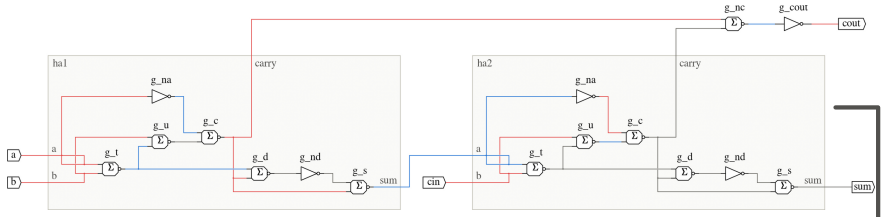
TritVerilog

```
module full_add(
  input  trit a, b, cin,
  output trit sum, cout);
  wire trit s1, c1, c2, nc;
  half_add ha1(.a(a), .b(b),
               .sum(s1), .carry(c1));
  half_add ha2(.a(s1), .b(cin),
               .sum(sum), .carry(c2));
  nany g_nc  (nc, c1, c2);
  inv  g_cout(cout, nc);
endmodule
```

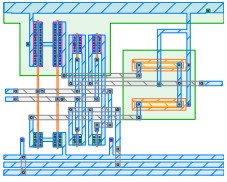
a full adder: a + b + cin = sum + 3·cout
built from two half adders

Synthesis



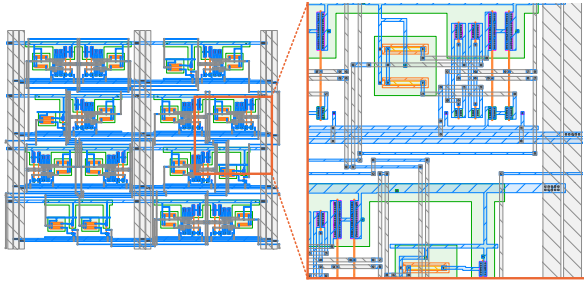
16 cells (11 NANY + 5 INV); at a = b = cin = +1:
1 + 1 + 1 = 0 + 3·(+1) (red +1, blue −1, gray 0)

Standard cell



NANY, from the taped-out
circuit; 3 rails

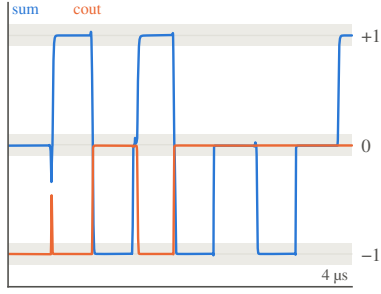
Place & route



876 × 702 µm, M1 + M2
DRC 0, LVS match

magnified: transistors,
M1, M2 and vias

Post-layout SPICE



extracted devices + wire C
three levels, ±0.5 V bands